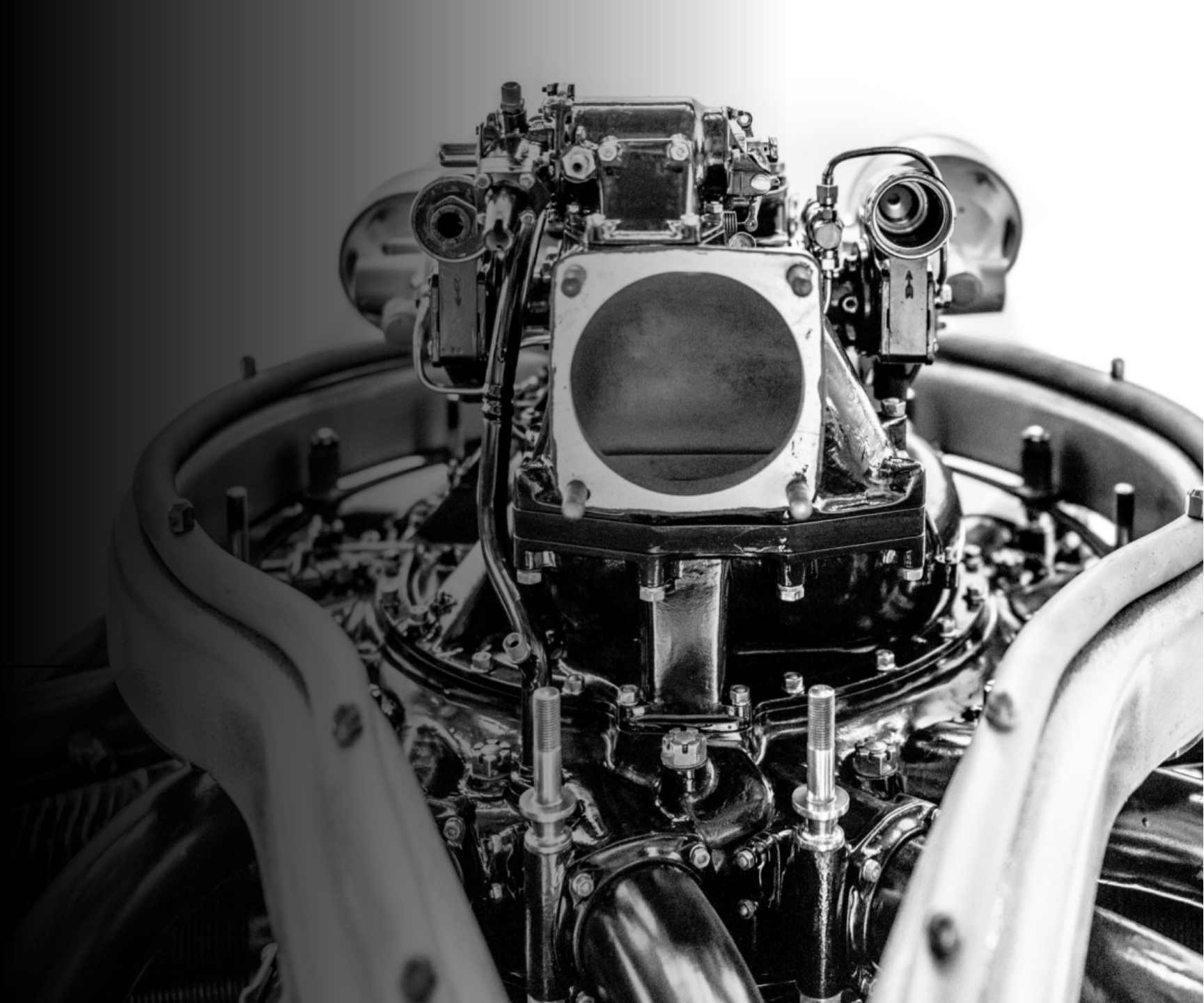




# ECE 575 project

ECE 575 – Computer Architecture  
Ashwin Subramanian and Shaurya Doger  
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Mr. Furat Al-Obaidy



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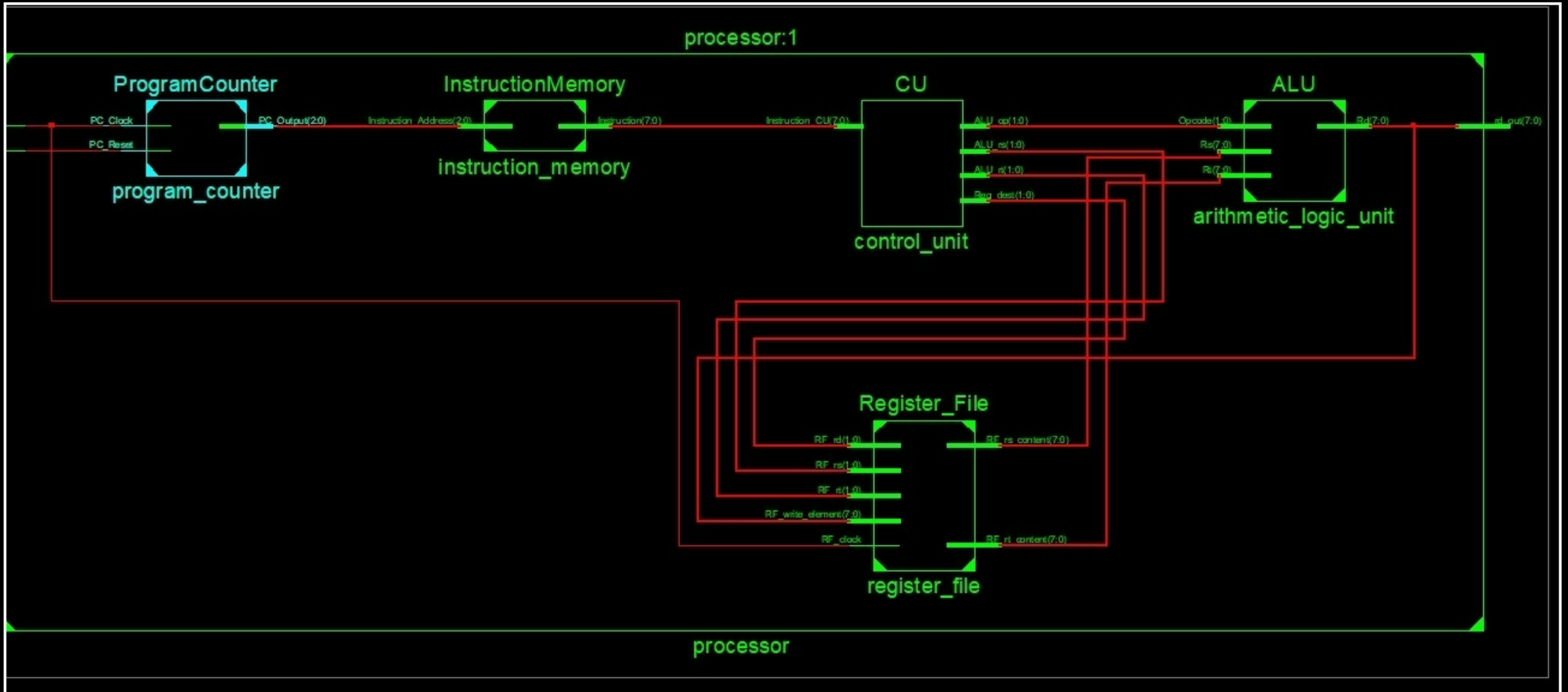
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# Introduction

- The microprocessor is a central processing unit (CPU) that serves as the brain of a computer, executing instructions and performing arithmetic/logic operations on data. It processes and controls the flow of data within a computer system.
- A microprocessor is an integrated circuit that contains the arithmetic, logic, control, and input/output (I/O) functions essential for executing instructions in a computer. It interprets instructions from memory, performs calculations, manages data flow, and coordinates various tasks, playing a pivotal role in enabling the functionalities of a computing device.

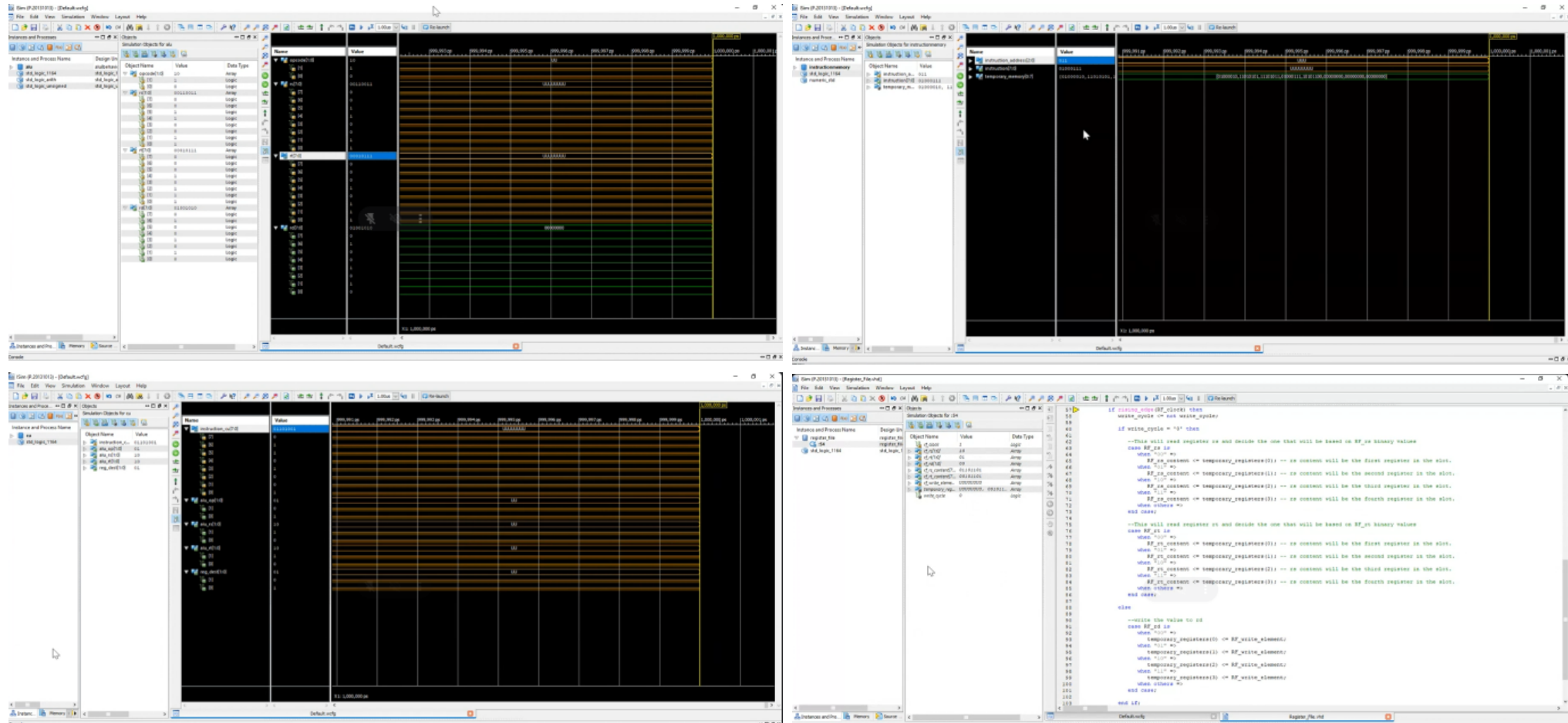
# Components used

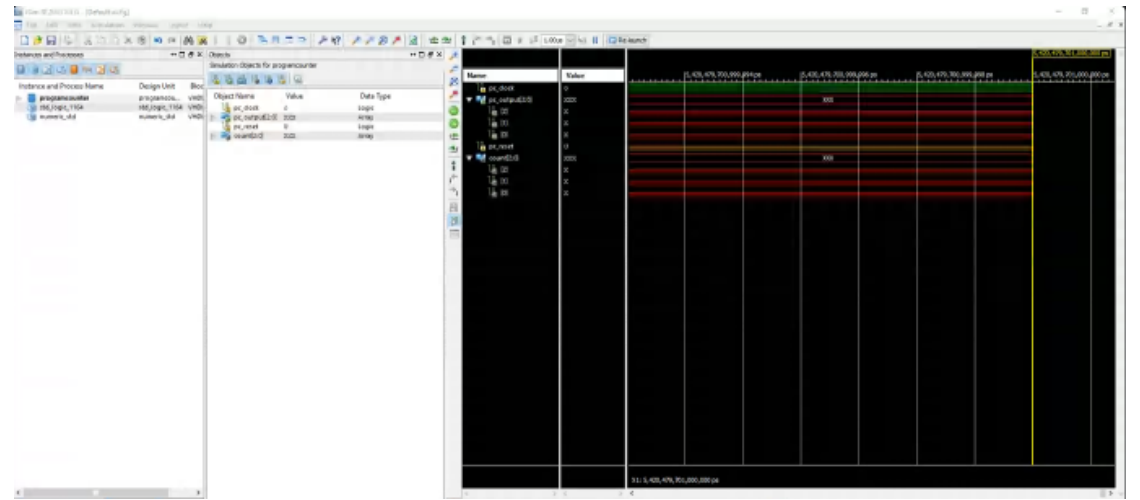
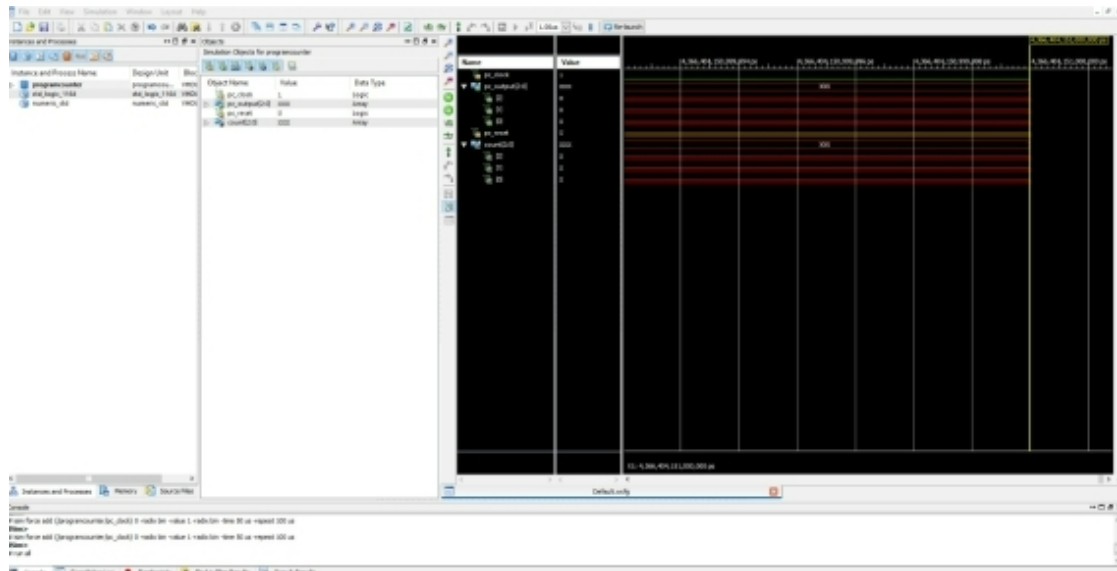
- The main components of a CPU are the program counter, the instruction memory, the ALU and the control unit.
- The program counter is a CPU register that keeps track of the memory address of the next instruction to be fetched and executed. It facilitates sequential instruction execution by incrementing to the subsequent memory location after each instruction.
- Instruction memory is a storage component in a CPU that holds the program instructions to be executed. It provides the CPU with the necessary instructions, fetched based on the address stored in the program counter, for processing and execution.
- The Arithmetic Logic Unit (ALU) is a CPU component performing arithmetic and logic operations on data. It executes computations like addition, subtraction, AND, OR, and other mathematical/logical functions within a computer processor.
- The control unit directs operations within a CPU, coordinating data flow and executing instructions. It interprets instructions, generates control signals, and coordinates the execution of tasks across various CPU components.



RTL Schematic

# VHDL Testbench Results





# Challenges Faced

- In the project, a key hurdle we encountered pertained to managing the clocking input. The output results did not come accurately as the clocking input did not work properly.